Math 248 Final Exam

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Contents

Problem 1. Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices in $M(n, \mathbb{C})$. Prove that

$$\operatorname{tr}(A^T B) = \sum_{i,j=1}^{n} a_{ij} b_{ij}.$$

Proof. Note that $(A^T)_{ij} = A_{ji}$ for all $1 \le i \le n$ and $1 \le j \le n$ and that

$$(A^T B)_{ij} = \sum_{k=1}^{n} A^T_{ik} B_{kj}$$

and by definition of trace, we have

$$\begin{aligned}
\operatorname{tr}(A^T B) &= \sum_{i=1}^{n} (A^T B)_{ii} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} A^T_{ij} B_{ji} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ji} B_{ji} \\
&= \sum_{i,j=1}^{n} A_{ji} B_{ji}.
\end{aligned}$$

By relabeling the indices, we obtain

$$\operatorname{tr}(A^T B) = \sum_{i,j=1}^{n} A_{ji} B_{ji}.$$

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Problem 2. Let $A, B \in M(n, \mathbb{C})$. Prove that

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

Proof. Note that $(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$ for all $1 \le i \le n$ and $1 \le j \le n$. Then we obtain

$$\begin{aligned}
\operatorname{tr}(AB) &= \sum_{i=1}^{n} (AB)_{ii} = \sum_{i=1}^{n} \left[\sum_{j=1}^{n} A_{ij} B_{ji} \right] \\
&= \sum_{i=1}^{n} \left[\sum_{j=1}^{n} B_{ji} A_{ij} \right] \\
&= \sum_{j=1}^{n} (BA)_{jj} \\
&= \operatorname{tr}(BA).
\end{aligned}$$

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Problem 3. Prove that $T_I(U(n)) = \operatorname{skew}(n, \mathbb{C})$.

Proof. Note that $U(n) = \{A \in M(n, \mathbb{C}) : A^*A = I\} \subseteq \mathrm{GL}(n, \mathbb{C})$.Our goal is to show that $T_I(U(n)) \subseteq \mathrm{skew}(n, \mathbb{C})$ and $\mathrm{skew}(n, \mathbb{C}) \subseteq T_I(U(n))$. Let $X \in T_I(U(N))$ be arbitrary. Then there exists a smooth path $A : (-\varepsilon, \varepsilon) \to U(n)$ such that $A(0) = I$ and $A'(0) = X$. Then for all $t \in (-\varepsilon, \varepsilon)$, we have $A(t) \in U(n)$ which by definition implies that

$$[A(t)]^* A(t) = I.$$

Differentiating both sides, we obtain

$$\frac{d}{dt}[A(t)^* A(t)] = 0 \implies A'(t)^* A(t) + A(t)^* A'(t) = 0.$$

In particular, for $t = 0$, we have

$$\begin{aligned} A'(0)^* A(0) + A(0)^* A'(0) = 0 &\implies X^* I + I^* X = 0 \\ &\implies X^* + X = 0 \\ &= X^* = -X. \end{aligned}$$

Hence, we have $X^* = -X$ and so we have $x \in \mathrm{skew}(n, \mathbb{C})$. Thus, we obtain our first containment.

On the other hand, let $X \in \mathrm{skew}(n, \mathbb{C})$. Define the path $A(t) = e^{tX}$. Clearly, we see that

$$A(0) = I \text{ and } A'(0) = X.$$

We need to show that

$$A(t)^* A(t) = I$$

in order for $A(t)$ to be a path in $U(n)$. Since $e^{X^*} = (e^X)^*$ and $(tX)^* = -tX$ for all $X \in M(n, \mathbb{C})$, we obtain

$$\begin{aligned} A(t)^* A(t) &= (e^{tX})^* (e^{tX}) \\ &= e^{(tX)^*} e^{tX} \\ &= e^{-tX} e^{tX} \\ &= e^{-tX + tX} \\ &= e^{O} \\ &= I. \end{aligned}$$

Thus, $X \in \mathrm{skew}(n, \mathbb{C})$ which gives is our second containment. ■

Problem 4. Let $A = (a_{ij}) \in M(2, \mathbb{C})$ be an upper triangular matrix. Prove that $\exp(A)$ is also upper triangular, and that the diagonal entries of $\exp(A)$ are $e^{a_{11}}$ and $e^{a_{22}}$.

Proof. Note that $A = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix}$. Once can easily show, via induction, that

$$A^n = \begin{pmatrix} a_{11}^n & a_{12}^n \\ 0 & a_{22}^n \end{pmatrix}.$$

Since $M(2,\mathbb{R}) \cong \mathbb{R}^4$, we can write

$$\begin{aligned}
\exp(A) = \sum_{k=1}^{\infty} \frac{A^k}{k!} &= \lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{k!} \begin{pmatrix} a_{11}^k & a_{12}^k \\ 0 & a_{22}^k \end{pmatrix} \\
&= \lim_{n\to\infty} \sum_{k=1}^{n} \begin{pmatrix} \frac{a_{11}^k}{k!} & a_{12}^k \\ 0 & \frac{a_{22}^k}{k!} \end{pmatrix} \\
&= \lim_{n\to\infty} \begin{pmatrix} \sum_{k=1}^{n} \frac{a_{11}^k}{k!} & a_{12}^k \\ 0 & \sum_{k=1}^{n} \frac{a_{22}^k}{k!} \end{pmatrix} \\
&= \begin{pmatrix} \sum_{k=1}^{\infty} \frac{a_{11}^k}{k!} & a_{12}^k \\ 0 & \sum_{k=1}^{\infty} \frac{a_{22}^k}{k!} \end{pmatrix}.
\end{aligned}$$

■

Problem 5. In HW3, you proved that

$$B^+(2,\mathbb{R}) = \text{the collection of all } 2\times 2 \text{ upper triangular matrices with positive diagonal entries}$$

is a subgroup of $\mathrm{GL}(2,\mathbb{R})$ that is topologically closed in $\mathrm{Gl}(2,\mathbb{R})$ (and hence a Lie subgroup of $\mathrm{GL}(2,\mathbb{R})$). Prove that

$$\mathrm{Lie}(B^+(2,\mathbb{R})) = \text{the space of all } 2\times 2 \text{ upper triangular matrices.}$$

Proof. Let X be an arbitrary upper triangular matrix. Define X in the following way:

$$X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} \\ 0 & x_{22} \end{pmatrix}.$$

Let $t \in \mathbb{R}$. Our goal is to show that $e^{tX} \in B^+(2,\mathbb{R})$. This will imply that $\mathfrak{g}$. From the previous problem, we have

$$e^{tX} = \begin{pmatrix} e^{tx_{11}} & x_{12}^n \\ 0 & e^{tx_{22}} \end{pmatrix}$$

where n is some integer. The above matrix is clearly upper triangular with positive diagonal entries. Hence, $e^{tX} \in B^+(2,\mathbb{R})$ and so $X \in \mathfrak{g} = \mathrm{Lie}(B^+(2,\mathbb{R}))$. Hence, we conclude that

$$\mathrm{Lie}(B^+(2,\mathbb{R})) = \text{the space of all } 2\times 2 \text{ upper triangular matrices.}$$

■

Problem 6. Let $F : M(n,\mathbb{R}) \to \mathbb{R}$ be defined by $F(A) = \det A$. Since $T_A M(n,\mathbb{R}) \equiv M(n,\mathbb{R})$ and $T_{F(A)}\mathbb{R} \equiv \mathbb{R}$ for each $A \in M(n,\mathbb{R})$, the derivative dF_A can be viewed as a linear map from $M(n,\mathbb{R})$ to $\mathbb{R}$. Prove that

$$\forall X \in M(n,\mathbb{R}) \quad dF_A = \mathrm{tr}(\mathrm{adj}(A)X).$$

Proof. Let $F(A) = \det(A)$ and let $X \in M(n,\mathbb{R})$. Let $\gamma : (-\varepsilon,\varepsilon) \to T_A M(n,\mathbb{R})$ such that $\gamma(0) = A$ and $\gamma'(0) = X$ and let $\beta : (-\varepsilon,\varepsilon) \to T_{F(A)}\mathbb{R}$ defined by $\beta(t) = F(\gamma(t))$. Since $\gamma(t)$ is a smooth path in $T_A M(n,\mathbb{R})$, it follows from the main properties of the exponential that

$$\frac{d}{dt}[\det(\gamma(t))] = (\det(\gamma(t)))\mathrm{tr}([\gamma^{-1}(t)\cdot\gamma'(t)]). \tag{†}$$

Evaluating † at $t = 0$, we obtain

$$\begin{aligned} DF_A X = \beta'(0) &= \frac{d}{dt}\Big|_{t=0} F(\gamma(t)) \\ &= \det(\gamma(0))\mathrm{tr}[\gamma^{-1}(0)\cdot\gamma'(0)] \\ &= \det(A)\mathrm{tr}[A^{-1}\cdot X] \\ &= \det(A)\mathrm{tr}\Big[[\det A]^{-1}\mathrm{adj}(A)\cdot X\Big] \\ &= \det(A)\cdot\det(A)^{-1}\mathrm{tr}[\mathrm{adj}(A)X] \\ &= \mathrm{tr}[\mathrm{adj}(A)X]. \end{aligned}$$

Thus, we conclude that

$$\forall X \in M(n,\mathbb{R}) \ \ dF_A = \mathrm{tr}(\mathrm{adj}(A)X).$$

■

Problem 7. Let $F : \mathrm{GL}(n,\mathbb{R}) \to M(n,\mathbb{R})$ be defined by $F(A) = A^{-1}$. Since $T_A\mathrm{GL}(n,\mathbb{R}) \equiv M(n,\mathbb{R})$ and $T_{F(A)}M(n,\mathbb{R}) \equiv M(n,\mathbb{R})$, for each $A \in \mathrm{GL}(n,\mathbb{R})$, the derivative dF_A can be viewed as a linear map from $M(n,\mathbb{R})$. Prove that

$$\forall X \in M(n,\mathbb{R}) \ \ dF_A(X) = -A^{-1}XA^{-1}.$$

Proof. Let $X \in M(n,\mathbb{R})$. Let $\gamma : (-\varepsilon,\varepsilon) \to T_A\mathrm{GL}(n,\mathbb{R})$ be a smooth path such that $\gamma(0) = A$ and $\gamma'(0) = X$ and let $\beta : (-\varepsilon,\varepsilon) \to T_{F(A)}M(n,\mathbb{R})$ be defiend by $F\circ\gamma$ i.e $\beta(t) = [\gamma(t)]^{-1}$. Since $\mathrm{GL}(n,\mathbb{R})$ is a matrix Lie group, we have $\beta'(t) = -\gamma^{-1}(t)\gamma'(t)\gamma^{-1}(t)$ where $DF_A(X) = \beta'(0)$. Hence, we have that

$$\beta'(0) = -\gamma^{-1}(0)\gamma'(0)\gamma^{-1}(0) = -A^{-1}XA^{-1}.$$

Hence,

$$DF_A(X) = -A^{-1}XA^{-1} \ \ \forall X \in M(n,\mathbb{R}).$$

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Problem 8. Let G_1 and G_2 be two matrix Lie groups, and let $F : G_1 \to G_2$ be a Lie group homomorphism. Let e denote the identity element of G_1, and let $dF_e : \mathfrak{g}_1 \to \mathfrak{g}_2$ be the induced Lie Algebra homomorphism. Prove that

$$\forall t \in \mathbb{R}, \forall X \in \mathfrak{g}_1 \ \ F(\exp_{G_1}(tX)) = \exp_{G_2}(tdF_e(X))$$

and, in particular,

$$\forall x \in \mathfrak{g}_1 \ \ F\circ\exp_{G_1} = \exp_{G_2}\circ dF_e(X).$$

Proof. Define $\gamma_X(t) = F(\exp_{G_1}(tX))$. To show that $F(\exp_{G_1}(tX)) = \exp_{G_2}(tdF_e(X))$, it suffices to show the following four properties:

(i) γ_X is smooth

(ii) For all $s,t \in \mathbb{R}$, $\gamma_X(s+t) = \gamma_X(s)\gamma_X(t)$

(iii) $\gamma_X(0) = I$

(iv) $\gamma'_X(0) = dF_e(X)$

(i) Since $F : G_1 \to G_2$ is a Lie group homomorphism, we know that F is also a smooth map. Clearly, $\exp_{G_1}$ is also a smooth map from $\mathfrak{g}_1 \to G_1$ and so their composition $F\circ\exp_{G_1}$ must also be a smooth map from $\mathfrak{g}_1 \to G_2$. Hence, γ_X is smooth.

(ii) Let $x \in \mathfrak{g}$ and let $s, t \in \mathbb{R}$. Since F is a Lie group homomorphism, we have that

$$\begin{aligned}(F \circ \exp_{G_1})\Big((s+t)X\Big) &= F\Big(\exp_{G_1}(sX + tX)\Big) \\ &= F\Big(\exp_{G_1}(sX)\exp_{G_1}(tX)\Big) \\ &= F(\exp_{G_1}(sX))F(\exp_{G_1}(tX)) \\ &= (F \circ \exp_{G_1}(sX))(F \circ \exp_{G_1}(tX)).\end{aligned}$$

Hence, we see that

$$(F \circ \exp_{G_1})\Big((t+s)X\Big) = (F \circ \exp_{G_1})(tX)(F \circ \exp_{G_1}(sX)).$$

Thus, we obtain

$$\gamma_X(t+s) = \gamma_X(t)\gamma_X(s).$$

(iii) Since F is a Lie group homomorphism, we have

$$\begin{aligned}\gamma_X(O) &= (F \circ \exp_{G_1})(OX) \\ &= F(\exp_{G_1}(O)) \\ &= F(I) \\ &= I.\end{aligned}$$

(iv) Observe that

$$\begin{aligned}\gamma'_X(O) &= \frac{d}{dt}\Big|_{t=0}\Big(F \circ \exp_{G_1}(tX)\Big) \\ &= \frac{d}{dt}\Big|_{t=0}F\Big(\exp_{G_1}(tX)\Big) \\ &= dF_e(X)\end{aligned}$$

where $\exp_{G_1}(tX)$ is a smooth path such that $\exp_{G_1}(O) = I$ and $\frac{d}{dt}\Big|_{t=0}\exp(tX) = X$. From properties (i) through (iv), it follows that $\gamma_X(t) = \exp_{G_2}(tX)$, and so we conclude that

$$F \circ \exp_{G_1}(tX) = \exp_{G_2}(tdF_e(X))$$

for all $t \in \mathbb{R}$ and $x \in \mathfrak{g}_1$. In particular, we have for $t = 1$, for all $x \in \mathfrak{g}_1$,

$$F \circ \exp_{G_1}(X) = \exp_{G_2}(dF_eX).$$

■

Problem 9. Let G_1 and G_2 be two matrix Lie groups, and let $F : G_1 \to G_2$ be a Lie group homomorphism. Let e denote the identity element of G_1. Prove that the differential $dF_e : \mathfrak{g}_1 \to \mathfrak{g}_2$ preserves the Lie Bracket; that is, prove that

$$\forall X, Y \in \mathfrak{g}_1 \quad dF_e([X, Y]) = [dF_eX, dF_eY].$$

Proof. Let $X, Y \in \mathfrak{g}_1$. Define $\gamma(s) = e^{tX}e^{sY}e^{-tX}$. Then we obtain

$$\begin{aligned}
[dF_eX, dF_eY] &= \frac{d}{dt}\Big|_{t=0}[e^{tdF_eX}dF_eYe^{-tdF_eX}] \\
&= \frac{d}{dt}\Big|_{t=0}\Big[F(e^{tX})dF_eYF(e^{-tX})\Big] && \text{(Naturality of Exponent Map)} \\
&= \frac{d}{dt}\Big|_{t=0}F(e^{tX})\frac{d}{ds}\Big(\Big|_{s=0}F(e^{sY})\Big)F(e^{-tX}) \\
&= \frac{d}{dt}\Big|_{t=0}\frac{d}{ds}\Big|_{s=0}F(e^{tX})F(e^{sY})F(e^{-tX}) && \text{(F is a Lie group homomorphism)} \\
&= \frac{d}{dt}\Big|_{t=0}\frac{d}{ds}\Big|_{s=0}F(e^{tX}e^{sY}e^{-tX}) \\
&= \frac{d}{dt}\Big|_{t=0}dF_e(e^{tX}Ye^{-tX}) \\
&= dF_e\Big(\frac{d}{dt}\Big|_{t=0}e^{tX}Ye^{-tX}\Big) && \text{(dF_e is a linear operator)} \\
&= dF_e([X, Y]).
\end{aligned}$$

Thus, we conclude that

$$[dF_eX, dF_eY] = dF_e([X, Y]).$$

■

Problem 10. Prove that $\mathrm{SL}(n, \mathbb{C})$ is path connected.

Proof. To show that $\mathrm{SL}(n, \mathbb{C})$ is path connected, we need to show that there exists a continuous curve connecting an arbitrary element $A \in \mathrm{GL}(n, \mathbb{C}^+)$ to the identity I. Indeed, define $\gamma : \mathrm{GL}(n, \mathbb{C}^+) \to \mathrm{SL}(n, \mathbb{C})$ by

$$A \longmapsto \begin{pmatrix} \frac{a_{11}}{\det A} & \cdots & \frac{a_{1n}}{\det A} \\ a_{21} & \ddots & \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}.$$

Note that for each $A \in \mathrm{GL}(n, \mathbb{C}^+)$, we have $\det f(A) = 1$ and so γ is well-defined. Also, note that γ is surjective. That is, for all $B \in \mathrm{SL}(n, \mathbb{C})$, $\gamma(B) = B$. Now, entries of $\gamma(B)$ are rational functions of entries of B (which are continuous). Since γ is

(1) well-defined

(2) surjective

(3) The entries of $\gamma(B)$ are entries of B

which are continuous functions as well. Now, we see that $\mathrm{SL}(n, \mathbb{C}) = \gamma(\mathrm{GL}(n, \mathbb{C}))$ is path-connected. ■

Theorem. The determinant of a block diagonal matrix is the product of the determinants of its diagonal blocks.

Theorem. Given a matrix $R \in \mathrm{SO}(n)$, we can decompose R as $R = AMA^T$, where $A \in O(n)$ and M is a block diagonal matrix with each block either a 2×2 rotation matrix or 1.

Problem 11. Suppose M is an $n \times n$ block diagonal matrix with each block either a 2×2 rotation matrix or 1.

(11-1) Prove that $M \in \mathrm{SO}(n)$.

Proof. Recall that $\mathrm{SO}(n) = \{A \in M(n, \mathbb{R}) : A^T A = I \text{ with } \det A = 1\}$. Suppose M is an $n \times n$ block diagonal matrix with each block either a 2×2 rotation matrix or 1; that is, for each M_i (block matrix along the diagonal of M) either

$$M_i = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix} \text{ or } M_i = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad (1 \leq i \leq n)$$

where $\theta_i \in \mathbb{R}$. In either case, we see that $\det M_i = 1$. Since the determinant of a block diagonal matrix is the product of determinants of each M_i, it follows that

$$\det M = \prod_{i=1}^{n} \det(M_i) = 1 \cdot 1 \cdot \ \cdots \ \cdot 1 = 1.$$

Also, notice that the transpose of M is just the transpose of its diagonal block matrices. If $M_i = I$, then clearly $M_i^T M_i = I$. If $M_i = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix}$, then

$$\begin{aligned} M_i^T M_i &= \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix} \begin{pmatrix} \cos\theta_i & \sin\theta_i \\ -\sin\theta_i & \cos\theta_i \end{pmatrix} \\ &= \begin{pmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \cos^2\theta + \sin^2\theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= I. \end{aligned}$$

Hence, for all $1 \leq i \leq n$, $M_i^T M_i = I$ and so $M^T M = I$. Thus, $M^T M = I$ and $\det M = 1$. Thus, $M \in \mathrm{SO}(n)$. ■

(11-2) Let $n \geq 2$. Prove that $\mathrm{SO}(n)$ is path connected.

Proof. Let $M \in \mathrm{SO}(n)$ as in $(11-1)$. To show that $\mathrm{SO}(n)$ is path connected, we need to find a continuous curve contained in $\mathrm{SO}(n)$ that connects M to the identity I. Indeed, define $\gamma : [0,1] \to \mathrm{SO}(n)$ by

$$\Gamma(t) = A\gamma(t)A^T$$

with

$$\gamma(t) = \begin{pmatrix} \gamma_1(t) & & & \\ & \gamma_2(t) & & \\ & & \ddots & \\ & & & \gamma_n(t) \end{pmatrix}$$

and $\gamma_i : [0,1] \to \mathrm{SO}(2)$ defined by

$$\gamma_i(t) = \begin{pmatrix} \sin(1-t)\theta_i & -\sin(1-t)\theta_i \\ \sin(1-t)\theta_i & \cos(1-t)\theta_i \end{pmatrix}. \qquad (1 \leq i \leq n)$$

Notice that each γ_i is continuous since each entry is clearly a continuous function and that $\Gamma(0) =$

M and $\Gamma(1) = I$ since for all $1 \leq i \leq n$ either

$$\gamma_i(0) = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix} \quad \text{or} \quad \gamma_i(1) = I.$$

Hence, it follows from the above that $\mathrm{SO}(n)$ is path-connected. ■

In your proof of the following problem, you may use the following fact:

Theorem. Given a symmetric matrix $A \in M(n, \mathbb{R})$, we can decompose A as $A = PDP^T$, where $P \in O(n)$ and D is a diagonal matrix whose diagonal entries are the eigenvalues of A.

Problem 12. Note that from (11-1) $M \in \mathrm{SO}(n)$ can be decomposed in the following way:

$$M = ARA^T$$

where $A \in O(n)$ and R is a block diagonal matrix with each block either 2×2 rotation matrix or 1.

(a) Explain why $\mathrm{SPD}(n, \mathbb{R})$ is a subset of $\mathrm{GL}(n, \mathbb{R})$.

Solution. Note that for every $A \in \mathrm{SPD}(n, \mathbb{R})$, A is diagonalizable; that is,

$$A = PDP^{-1}$$

where P is invertible and D is diagonal matrix with entries being the eigenvalues $\lambda_1, \ldots, \lambda_n$ of A. Since A is symmetric, we have $\lambda_i \in \mathbb{R}$ for all $1 \leq i \leq n$. Also, $\lambda_i > 0$ because A is positive-definite. Hence,

$$\begin{aligned} \det A &= \det(PDP^{-1}) \\ &= \det(D) \\ &= \prod_{i=1}^{n} \gamma_i \\ &> 0. \end{aligned}$$

Thus, $\det A > 0$ and so $A \in \mathrm{GL}(n, \mathbb{R})$. Therefore, $\mathrm{SPD}(n, \mathbb{R}) \subseteq \mathrm{GL}(n, \mathbb{R})$. ■

(b) $\mathrm{SPD}(n, \mathbb{R})$ is NOT a subgroup of $\mathrm{GL}(n, \mathbb{R})$.

Proof. We will show that $\mathrm{SPD}(n, \mathbb{R})$ is not closed under multiplication. Consider the following 2×2 matrices $A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$ and $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = B$. It is immediate that both A and B are symmetric positive definite. Now, observe that

$$\begin{aligned} AB &= \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} -1 & 3 \\ -3 & 8 \end{pmatrix}. \end{aligned}$$

Notice that $\det(AB) < 0$ and AB is clearly not symmetric. Hence, $AB \notin \mathrm{SPD}(n, \mathbb{R})$ and so $\mathrm{SPD}(n, \mathbb{R})$ is not closed under matrix multiplication. ■

(c) Prove that $\mathrm{SPD}(n, \mathbb{R})$ is a path-connected subset of $\mathrm{GL}(n, \mathbb{R})$.

Proof. Let $A \in \mathrm{SPD}(n, \mathbb{R})$. With the theorem given, we can decompose A as

$$A = PDP^T$$

where $P \in O(n)$ and D is diagonal with the diagonal entries are the eivenvalues of A. Define $\Gamma : [0,1] \to \mathrm{SPD}(n, \mathbb{R})$ by

$$\gamma(t) = P \begin{pmatrix} (1-t)\lambda_1 + t & & & \\ & (1-t)\lambda_2 + t & & \\ & & \ddots & \\ & & & (1-t)\lambda_n + t \end{pmatrix} P^{-1} \qquad (\forall t \in [0,1])$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A ($\lambda_i > 0 \in \mathbb{R}$ since $A \in \mathrm{SPD}(n, \mathbb{R})$) each diagonal entry of $\Gamma(t)$ is a continuous function. Hence, $\Gamma(t)$ is a continuous curve connecting $A \in \mathrm{SPD}(n, \mathbb{R})$ to $I \in \mathrm{SPD}(n, \mathbb{R})$. ■

You may use the following fact to prove the following problem.

Theorem. Given a matrix $A \in \mathrm{GL}(n, \mathbb{R})^+$, we can decompose A as $A = SQ$ where $S \in \mathrm{SPD}(n, \mathbb{R})$ and $Q \in \mathrm{SO}(n)$.

Problem 13. Prove $G = \mathrm{GL}(n, \mathbb{R})^+$ is path-connected.

Proof. Let $A \in \mathrm{GL}(n, \mathbb{R})^+$. Then by the given theorem, $A = SQ$ with $S \in \mathrm{SPD}(n, \mathbb{R})$ and $Q \in \mathrm{SO}(n)$. Since $S \in \mathrm{SPD}(n, \mathbb{R})$, we can decompose S as $S = PDP^T$ where $P \in O(n)$ and D is a diagonal matrix whose diagonal entries are the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ of S. Since $\mathrm{SPD}(n, \mathbb{R})$ is path, we can connect S to I by the following continuous curve $\Gamma_1 : [0,1] \to \mathrm{SPD}(n, \mathbb{R})$ defined by

$$\forall t \in [0,1] \quad \Gamma_1(t) = P \begin{pmatrix} (1-t)\lambda_1 + t & & & \\ & (1-t)\lambda_2 + t & & \\ & & \ddots & \\ & & & (1-t)\lambda_n + t \end{pmatrix} P^T.$$

Notice that with this curve $\Gamma_1(0) = S$ and $\Gamma_1(1) = I$. Likewise, we can decompose Q as $Q = AMA^T$ with $A \in O(n)$ and M is a block diagonal matrix with each block either a 2×2 rotation matrix or 1. Since $\mathrm{SO}(n)$ is path-connected, we can define a continuous curve $\Gamma_2; [0,1] \to \mathrm{SO}(n)$ by

$$\forall t \in [0,1] \quad \Gamma_2(t) = A\gamma(t) = \begin{pmatrix} \gamma_1(t) & & & \\ & \gamma_2(t) & & \\ & & \ddots & \\ & & & \gamma_n(t) \end{pmatrix} A^T$$

with some of the diagonal entries being

$$\gamma_i(t) = \begin{pmatrix} \cos(1-t)\theta_i & -\sin(1-t)\theta_i \\ \sin(1-t)\theta_i & \cos(1-t)\theta_i \end{pmatrix}$$

and some being equal to 1. Notice further that $\Gamma_2(0) = Q$ and $\Gamma_2(1) = I$. Now, notice further that $\Gamma_1\Gamma_2(0) = SQ = A$ and $\Gamma_1\Gamma_2(0) = I$. Also, since Γ_1 and Γ_2 are continuous, it follows that $\Gamma_1\Gamma_2$ is also continuous (where $\Gamma : [0,1] \to \mathrm{GL}(n, \mathbb{R})^+$ defined by $\Gamma(t) = \Gamma_1\Gamma_2(t)$ for all $t \in [0,1]$). Hence, we see that $\mathrm{GL}(n, \mathbb{R})^+$ is path-connected. ■